

Use AI to generate images (compatible with editing in PPT)

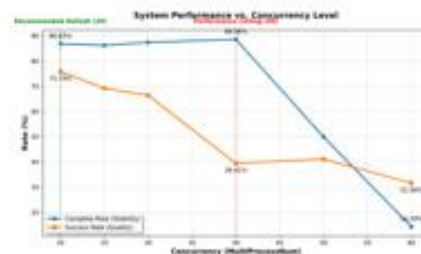
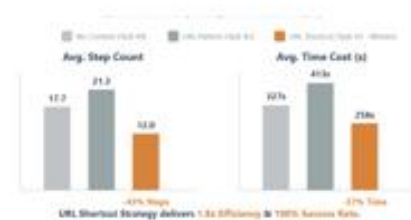
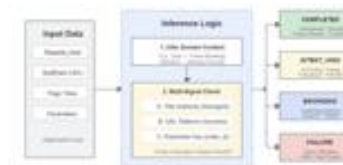
Lifang Zhou

Agenda

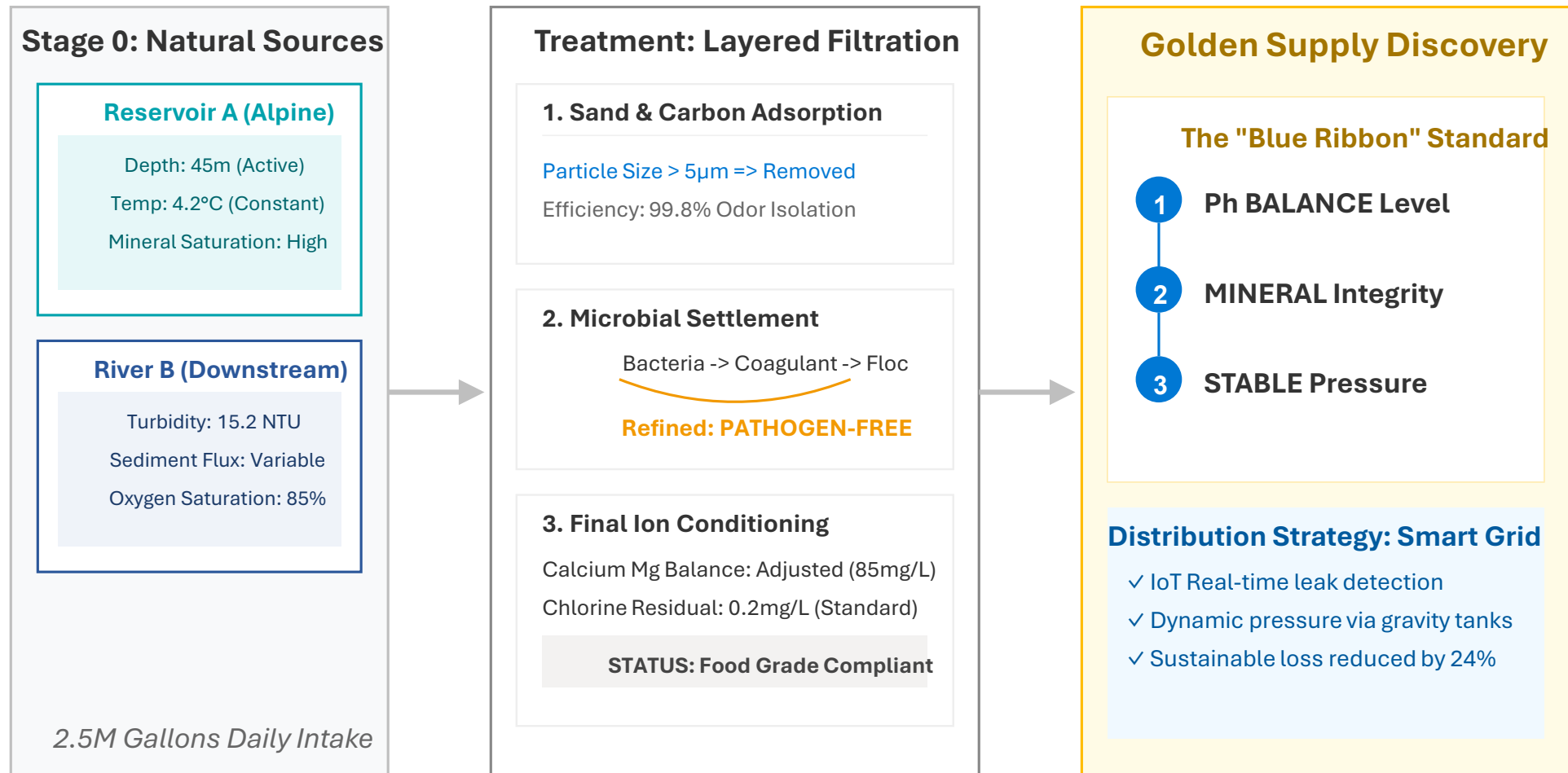
- Image cases by AI cost only in mins
- Quick walk through & have a try
- My best practice
- Pitfalls & Prompt (see slides 14 & 16 PPT Notes)
- QnA

images generated by AI in mins

- Most used in work report that including but not limited to flowcharts, data metrics charts, etc.

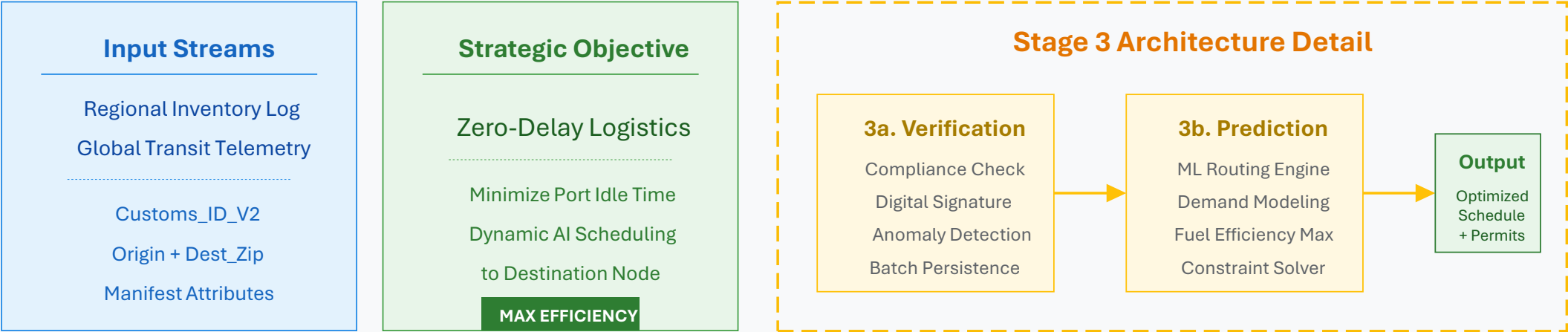
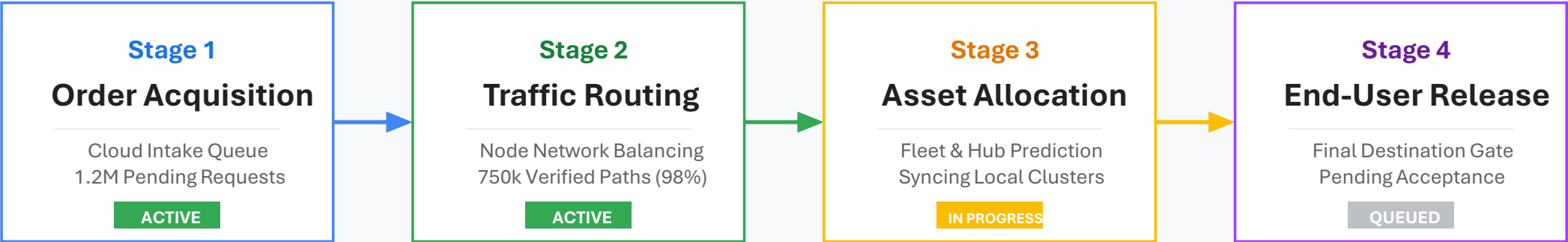


Work data anonymized with different content mocked by LLM,
but keep layout same of my cases



Global Supply Chain Logistics

Real-time Optimization and Autonomous Freight Management



Key Performance Indicators (KPI)				
1,245,800 Total Units	99.4% Fulfillment Rate	482 Active Hubs	84.2% Fleet Utilization	US/CN/DE Top Markets

Supply Chain Pipeline - Optimizing Global Logistics with Autonomous Network Orchestration | Data Source: Multi-modal Logistics API

Quick walk through & have a try

query_rewrite_diagram.svg

query_rewrite_diagram.svg

Modern Query Rewrite Pipeline

```
graph TD; A[User Query] --> B[Intent Classification<br/>(Needs Rewrite? Ambiguity?)]; B --> C[Generative AI (LLM)<br/>HyDE / Expansion<br/>Step-Back Prompting]; B --> D[Semantic Linking<br/>Vector Q2Q Similarity<br/>Dense Encoders]; B --> E[Rule-Based / Stats<br/>Synonym Mapping<br/>Search Logs (Q2Q)]; C --> F[Ranker & Scorer<br/>Collision Check<br/>Relevance Scoring (Cross-Encoder)]; D --> F; E --> F; F --> G[Optimized Query];
```

The diagram illustrates a 'Modern Query Rewrite Pipeline'. It starts with a 'User Query' box, which leads to an 'Intent Classification' box (labeled 'Needs Rewrite? Ambiguity?'). From 'Intent Classification', the flow splits into three parallel paths: 'Generative AI (LLM)' (labeled 'HyDE / Expansion' and 'Step-Back Prompting'), 'Semantic Linking' (labeled 'Vector Q2Q Similarity' and 'Dense Encoders'), and 'Rule-Based / Stats' (labeled 'Synonym Mapping' and 'Search Logs (Q2Q)'). All three paths converge into a 'Ranker & Scorer' box (labeled 'Collision Check' and 'Relevance Scoring (Cross-Encoder)'). Finally, the output is an 'Optimized Query' box.

CHAT

ROO CODE

MS ROO CODE

INDUSTRY APPROACHES TO QUERY REWRITE TECHNIQUES

rewritten candidates to select the one most likely to improve retrieval metrics (NDCG, MRR) without introducing "drift" (changing the user's original meaning).

okay, please just write a svg image to let me better learn this query rewrite tech. also I need to continue edit it in PPT, make sure it's compatible

svg_prompt.md

Created query_rewrite_diagram.svg

I've created a clean, modular SVG diagram for you at query_rewrite_diagram.svg .

How to use this in PowerPoint:

1. **Insert:** Go to **Insert > Pictures > This Device** and select query_rewrite_diagram.svg .
2. **Convert to Shapes:** Once inserted, right-click the image and select **"Convert to Shape"**. This will break the SVG into standard PowerPoint rectangles, text boxes, and lines that you can recolor, resize, or edit individually.
3. **Ungroup:** If it's still grouped, press **Ctrl + Shift + G** to ungroup the elements completely.

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

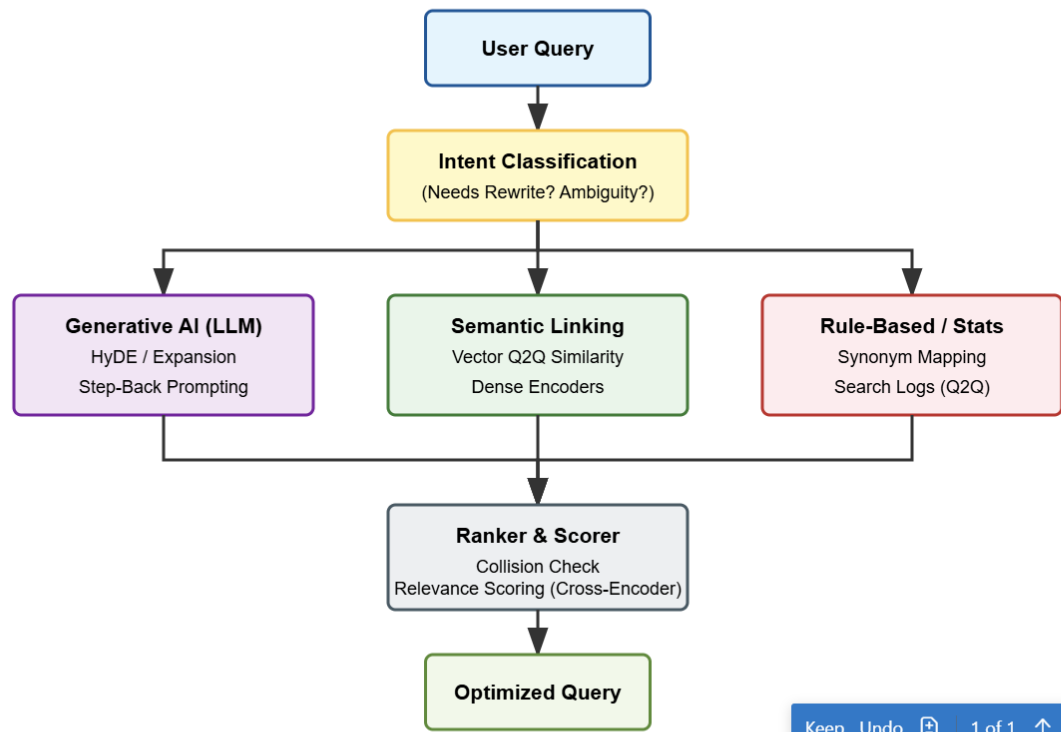
PORTS

AIProductivity>

powershell

1 of 1

Modern Query Rewrite Pipeline



PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

AIProductivity>

Tips:

1. recommend VL capacity model so you can have image as input
2. Install SVG Preview extension in VS Code
3. Prompt: **generate an image of SVG file, make sure that I can convert to shape then edit in PPT**
4. Must use PPT desktop

INDUSTRY APPROACHES TO QUERY REWRITE TECHNIQUES

rewritten candidates to select the one most likely to improve retrieval metrics (NDCG, MRR) without introducing "drift" (changing the user's original meaning).

Step 1

okay, please just write a svg image to let me better learn this query rewrite tech. also I need to continue edit it in PPT, make sure it's compatible

svg_prompt.md

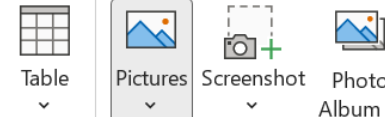
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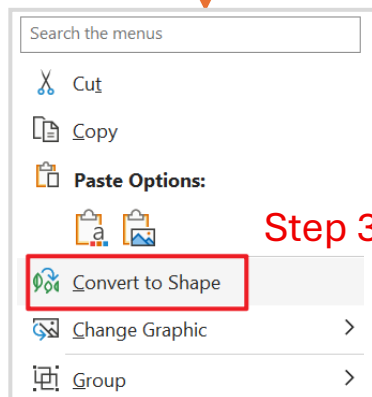
Insert Draw Design Transitions



Insert Picture From

- Brand Images...
- This Device...**
- Generate an Image...
- Stock Images...
- Online Pictures...

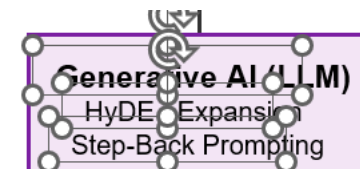
Step 2



Step 3

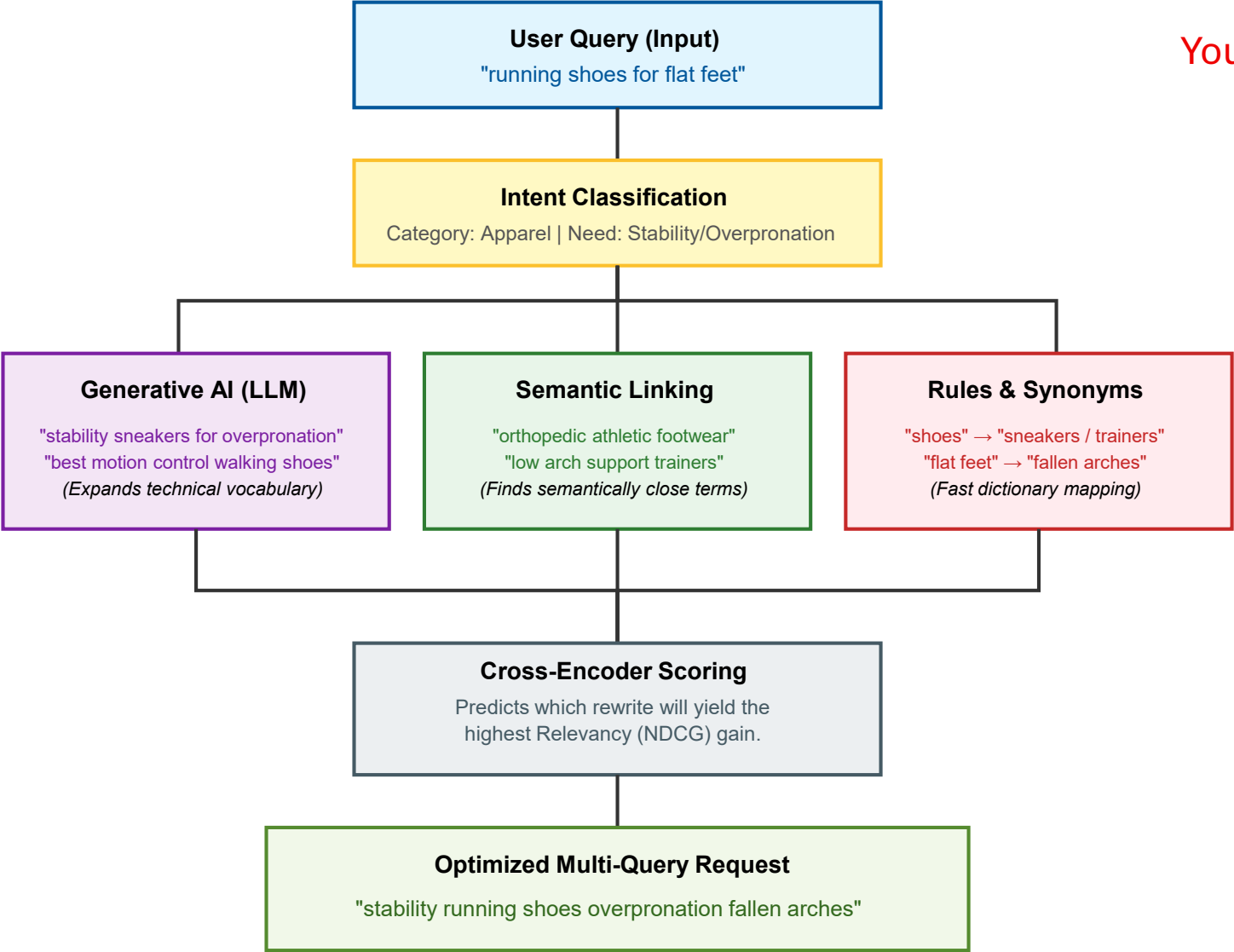
Can be modified freely at this time.

Step 4



Query Rewrite Example: User Intent to Retrieval

You can change everything here of this image



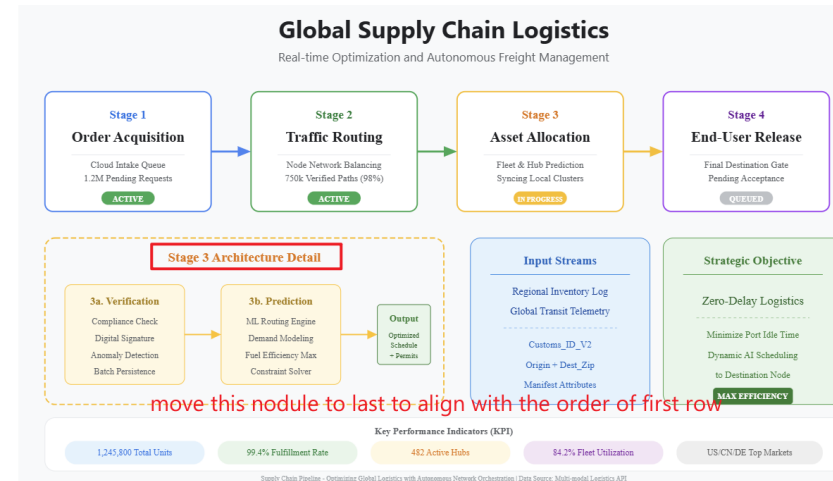
My best practice

Fell free to use multi-modal input, like Gemini 3, Claude

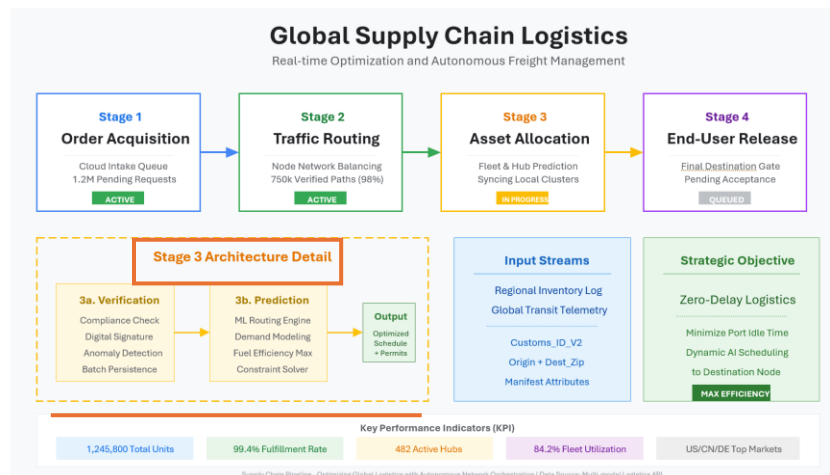
follow the instructions I commented in this snapshot, the original file is here
C:\Users\lifangzhou\OneDrive -
Microsoft\Desktop\AIProducuity\QuickDemo\logistics_pipeline_anon.svg

The input could be image

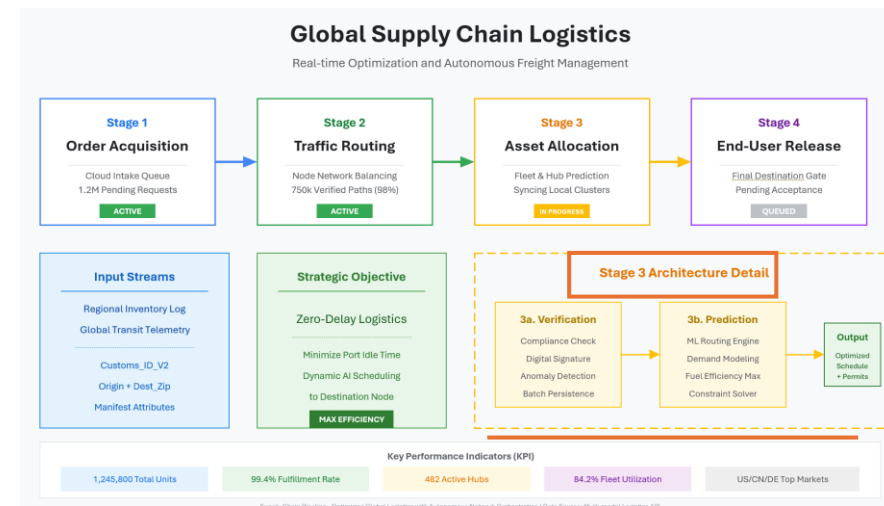
Pasted Image



move this nodule to last to align with the order of first row



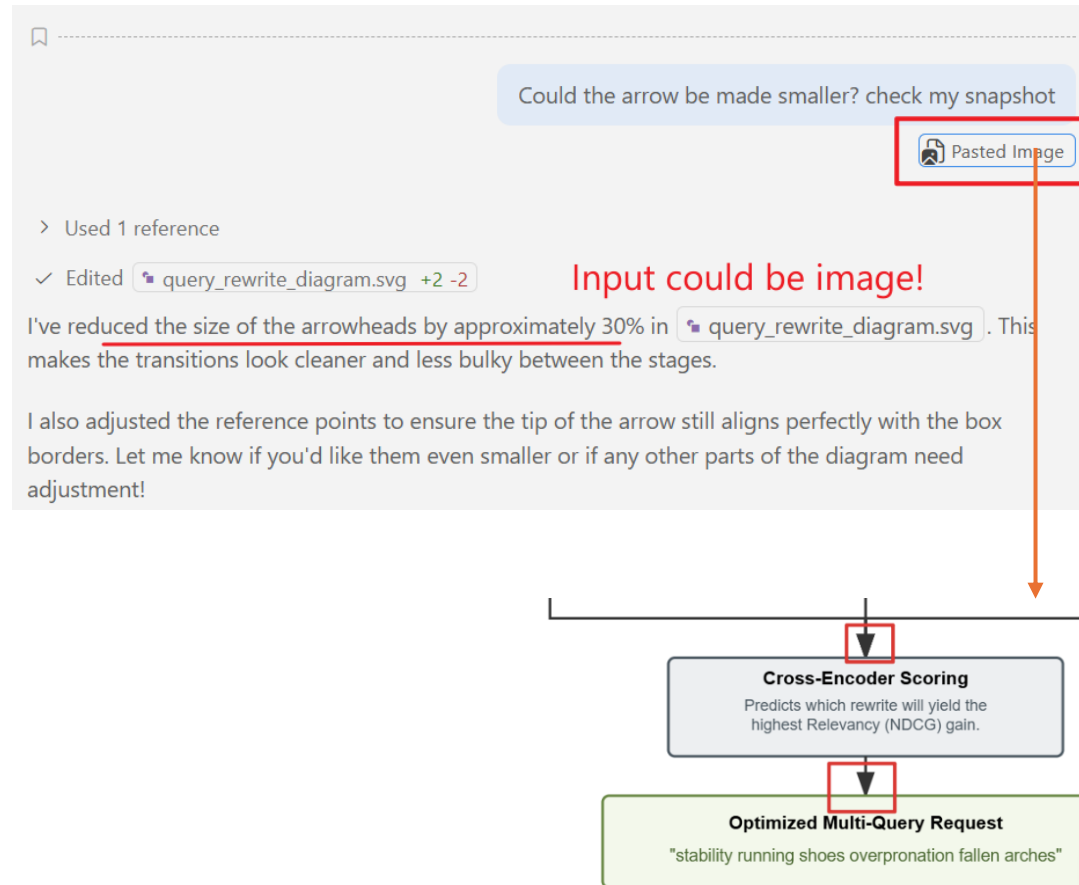
Before



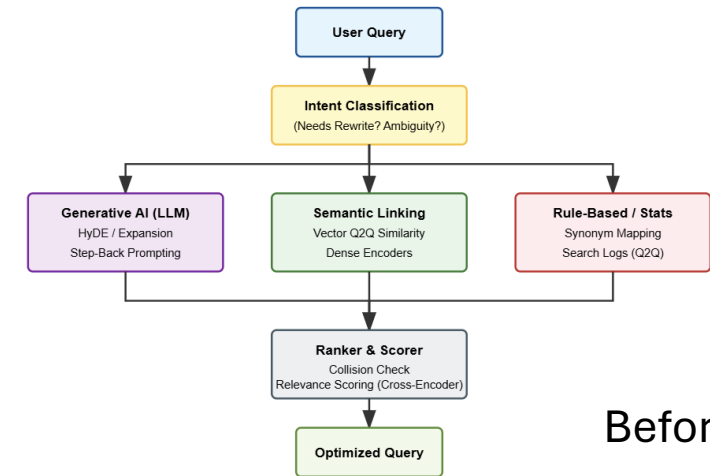
After

My best practice

Fell free to use multi-modal input using multi-modal LLM

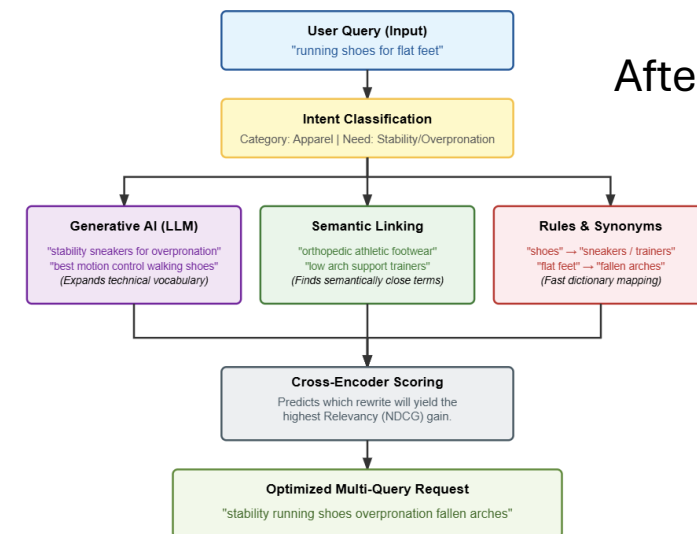


Modern Query Rewrite Pipeline



Before

Query Rewrite Example: User Intent to Retrieval



After

Multi modal do understand your multi-modal inputs

Pasted Image

My best practice

Fell free to use multi-modal input, Gemini 3 is good with it

Could the arrow be made smaller? check my snapshot

Used 1 reference

Edited query_rewrite_diagram.svg +2 -2

Input could be image!

I've reduced the size of the arrowheads by approximately 30% in query_rewrite_diagram.svg. This makes the transitions look cleaner and less bulky between the stages.

I also adjusted the reference points to ensure the tip of the arrow still aligns perfectly with the box borders. Let me know if you'd like them even smaller or if any other parts of the diagram need adjustment!

Before

Modern Query Rewrite Pipeline

User Query

Intent Classification (Needs Rewrite? Ambiguity?)

Generative AI (LLM) (Hybrid - Expansion, Step-Back Prompting)

Semantic Linking (Vector G2G Similarity, Dense Encoders)

Rule-Based / Stats (Recursion Mapping, Search Logic (G2G))

Ranker & Scorer (Custom Check, Relevance Scoring, Cross-Encoder)

Optimized Query

After

Query Rewrite Example: User Intent to Retrieval

User Query (Input) "running shoes for hot feet"

Intent Classification (Category: Apparel / Need: Stability/Comfortation)

Generative AI (LLM) "Stability needed for overpronation? User needs comfort walking on hot? (expand technical vocabulary)"

Semantic Linking "Stability: athletic footwear" "Hot feet: cushioned insole" "User wants support/comfort" (link: stability -> overpronation)

Rules & Synonyms "Stable" -> "Stability" "Comfort" "Hot feet" -> "Warm insole" (Final delivery mapping)

Cross-Encoder Scoring (Predicts which rewrite will yield the highest Relevance (NDCG@1) gain)

Optimized Multi-Query Request "stability running shoes overpronation fallen arches"

describe what my snapshot content is

Pasted Image

> Used 1 reference

✓ Optimizing tool selection...

Your snapshot serves as a **"Best Practice" guide for multi-modal AI interaction**, specifically showcasing how **Gemini 3 Flash** handles visual feedback to iterate on design.

The content is organized as follows:

- Workflow Demonstration:** On the left, it shows a chat interface where a user attaches an image and gives a visual-context instruction ("Could the arrow be made smaller?"). The AI successfully processes the image and edits the SVG code.
- Comparison of Iterations:** On the right, it contrasts two versions of the "Query Rewrite" pipeline:
 - "Before":** The initial generic version titled "Modern Query Rewrite Pipeline."
 - "After":** The refined version titled "Query Rewrite Example: User Intent to Retrieval," which incorporates specific data (the "running shoes" example) and the visually smaller arrowheads requested.
- Key Message:** The slide highlights that **"Input could be image!"** and encourages users to leverage multi-modal capabilities for precise UI/UX or diagram edits, noting that Gemini 3 is highly capable of bridging the gap between a visual request and code implementation.

My best practice 0 changes made

here I want to have a chart to demo the performance of different tasks, help me generate a svg file

actually these 3 task is the same task but with different configs, taskId is the config setting

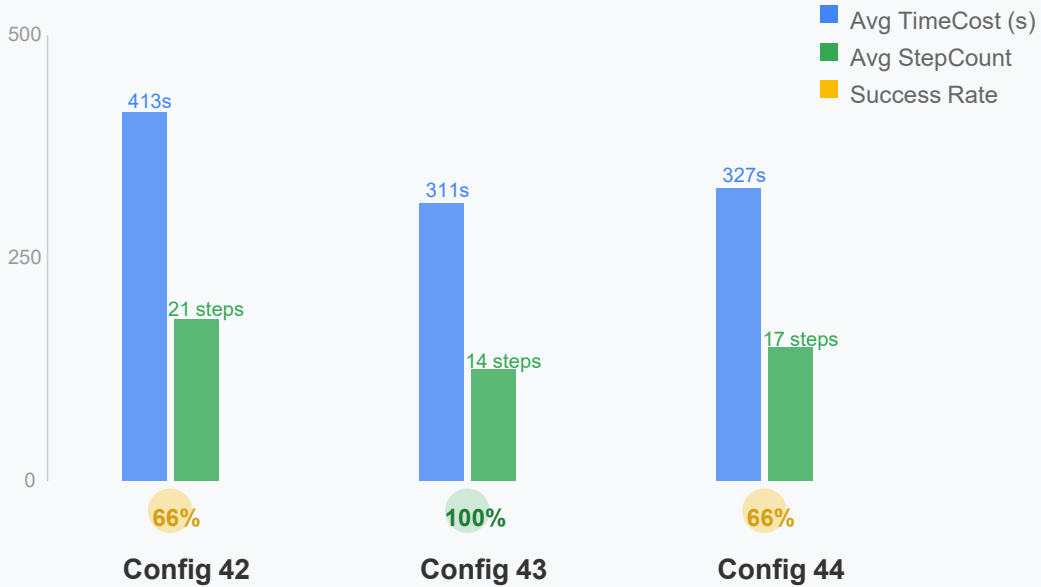
Pasted Image

Row Labels	Average of TimeCost	Average of StepCount
task_42	413.6496254	21.33333333
task_43	310.8812147	14.66666667
task_44	327.497901	17.66666667
Grand Total	350.676247	17.88888889

Row Labels	Count of EvalResult
task_42	3
failure	1
success	2
task_43	3
success	3
task_44	3
failure	1
success	2
Grand Total	9

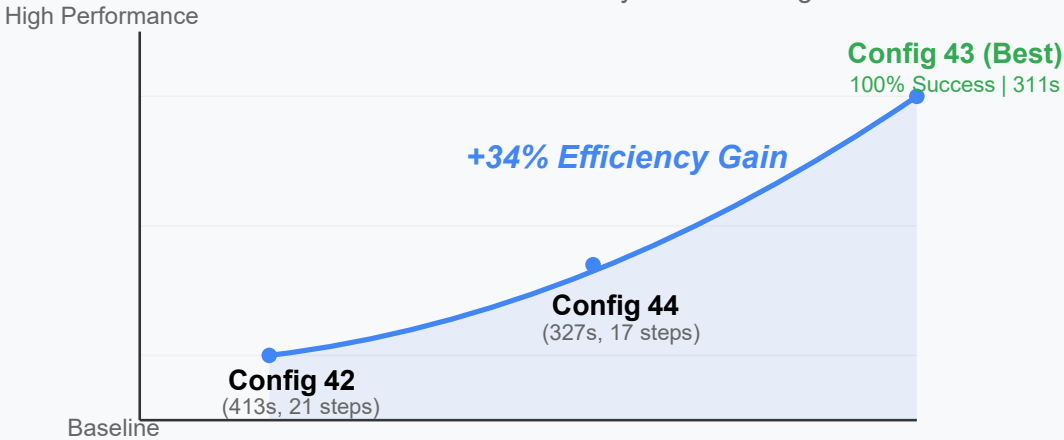
I want to show a Upward trend Curve graph in my work report, change this image

Configuration Performance Comparison



Model Optimization Trend

Performance Efficiency across Configurations



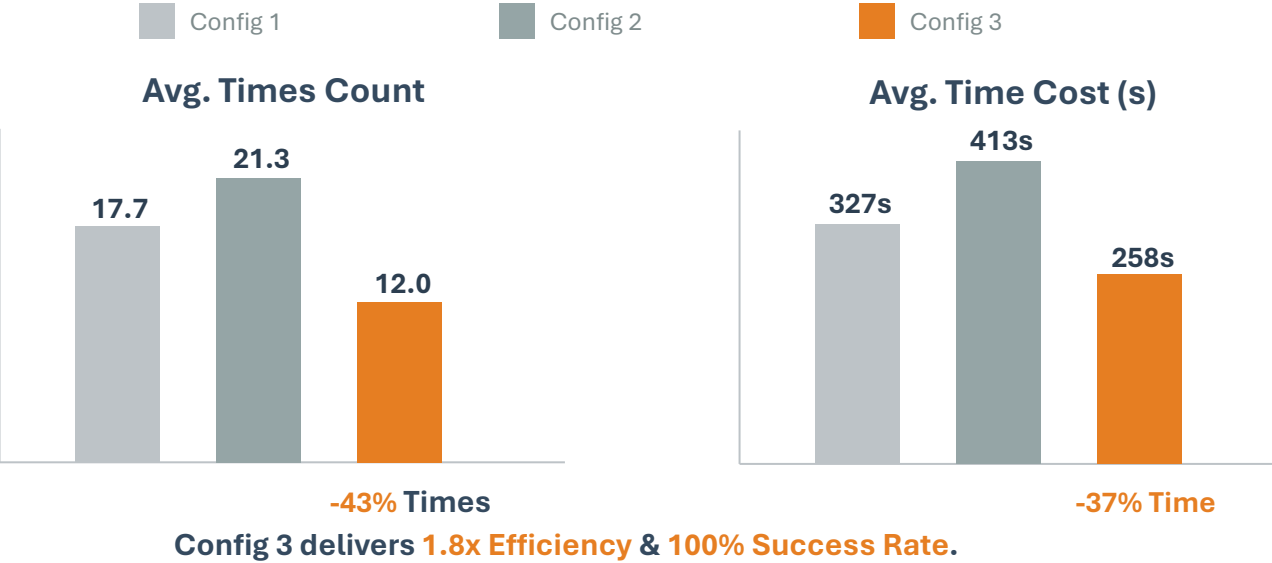
Pipeline Evolution

Input: a table snapshot image with 4 rows * 5 cols.
Prompt: based this data in the snapshot, show a up-forward trend SVG img to my leader



XX Optimization: Performance Benchmark

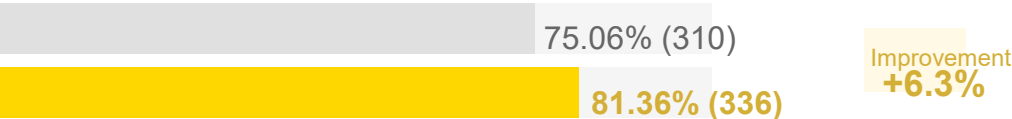
Impact of different XX Strategies on the same Task



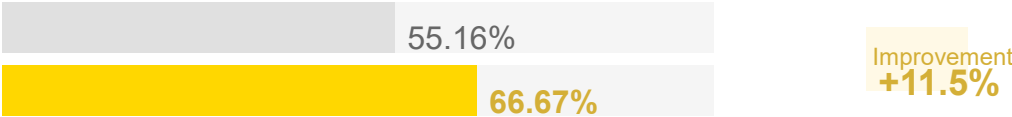
XX Mechanism Performance Gain

Total Tasks Analyzed: 413

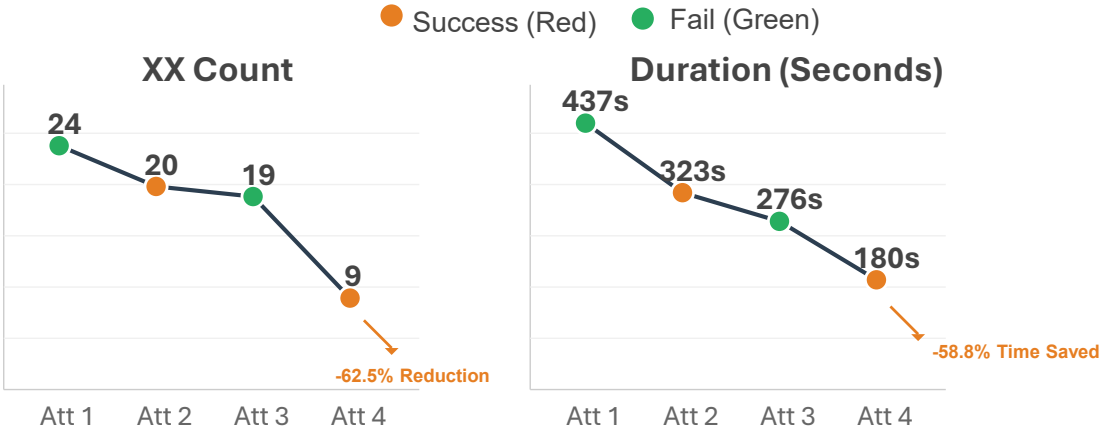
Resolved Tasks (Success Rate)



Stable Success Rate (Maintain Success)



XX Performance about YY



Pipeline efficiency significantly improved across 4 iterations.

Pro-level Compatibility Tactics

- Given enough context, you can generate PPT slides in SVG file seq

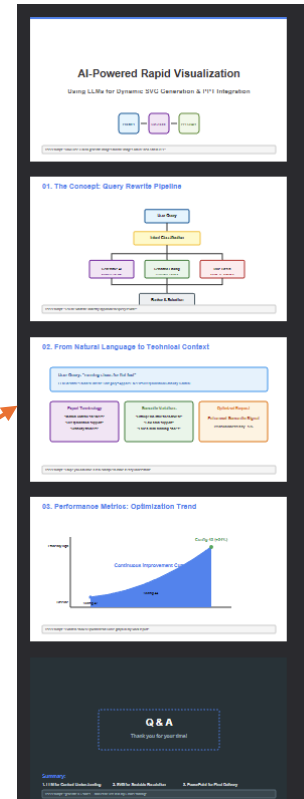
actually all the questions I ask you, is the demo case I want to show in a sharing that
how I use LLM to generate images and the images can be also edit in PPT

why not you just generate a 5 slides in C:\Users\lifangzhou\OneDrive - Microsoft\Desktop\AIProductivity\PPT_Slides
and let me use it in my 5 mins sharing , all the images is in svg format , and I will continue edit then in PPT ,

✓ PPT_Slides
├─ slide_1_intro.svg
├─ slide_2_concept.svg
├─ slide_3_example.svg
├─ slide_4_data.svg
└─ slide_5_qa.svg

concat all the svg file into a PDF orderly

A quick try without any edit



✓ url_pattern_PPT
> .venv
├─ 01_Title_Slide.svg
├─ 02_Pipeline_Overview.svg
├─ 03_Semantic_De_noising.svg
├─ 04_Quality_Audit_Logic.svg
├─ 05_Data_Yield_Funnel.svg
└─ 06_Summary_Impact.svg



A good example in my work,
enough context with 0 edits, accepted

Best Practices: *Clear logic leads to success diagrams.*

- **Context & verification is the key to precision**

- Context is King
- Verify LLM understand correct, Let it explain or summary

- **The "Summary-Detail" Principle**

- Don't cram everything into one sub-diagram.
- Use a "Total-to-Part" structure for clarity.

- **Describe Intent, Not Just Geometry**

- Define the 'Vibe': "Highlight upward trajectory."
- 80/20 Rule: AI build bones; You can skin the final 20%.
- **Other Specific requirements :**
"This is for a Tech Leader, I want Morandi..."

Specific Prompt Example:

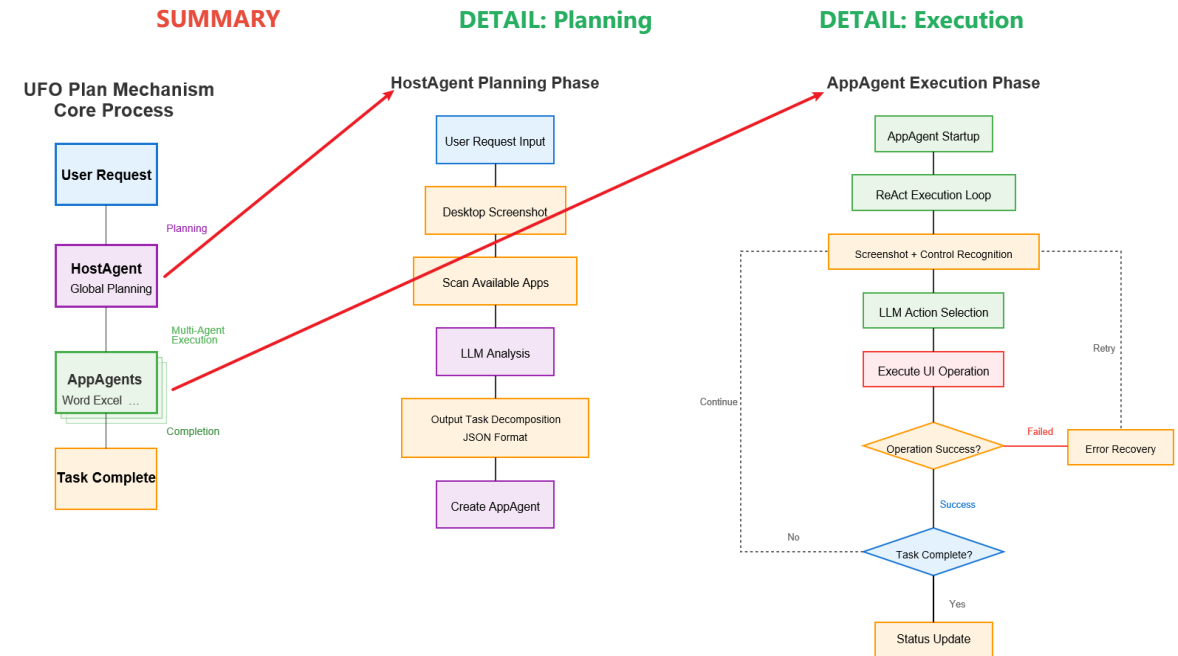
"Logic First: Emphasize pipeline capabilities."

"Style: using Morandi color palette,
with a professional sense of modern business/tech reports."

"Format: generate as SVG file, support PPT 'Convert to Shape'.
and follow guidance of the prompt file

"Audience: Sr tech lead who likes simple and direct"

"Divide and conquer."



"Logic First: Emphasize pipeline capabilities."

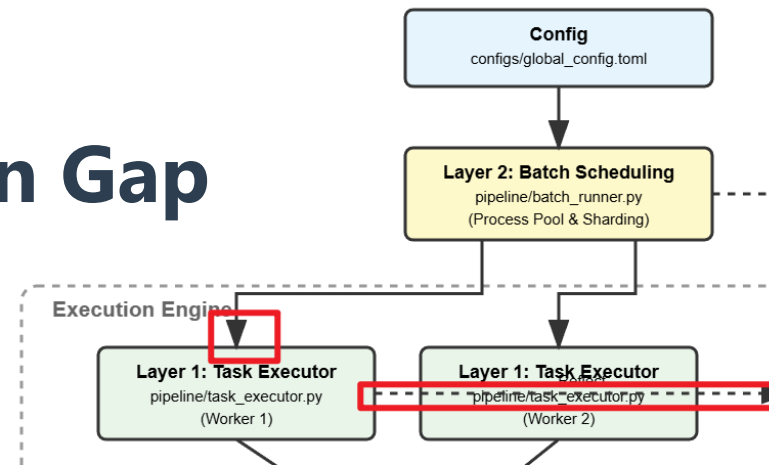
"Style: using Morandi color palette, with a professional sense of modern business/tech reports."

"Format: generate as SVG file, support PPT 'Convert to Shape'. and follow guidance of the prompt file"

"Audience: Sr tech lead who likes simple and direct"

This Slide is almost generated by LLM in SVG format

Pitfalls: The PPT Translation Gap



The Editing Friction (Standard SVG)

- 1. **The "Broken Bond" Arrow**
Moving a box doesn't pull the arrow.
Static polygons vs. Native PPT connectors.
- 2. **Layout Overlap & Drift**
Shapes often shift after "Convert to Shape."
Clean boundaries turn into messy overlaps.
- 3. **The "Flat Layer" Explosion**
Hierarchy is lost on import. You get 1000 tiny pieces instead of logical groups.

The Engineering Protocol (whole prompt in next slide)

- A. Arrow Atomicity**
Forced overlap (2-4px) between lines and polygon tips to eliminate rendering gaps.
- B. Recursive Grouping**
Deeply nested <g> tags (Main > Sub > Atom) allowing stepwise "Professional Ungrouping."
- C. Text Anchoring**
Mandatory text-anchor: middle and 1.2x width buffer to prevent font overflow.